

(a) Take $\mathbf{x}$ to be the direction to the right, $\mathbf{y}$ to be the upwards direction, and $\mathbf{z}$ is then out of the page. The initial velocity is in the $\mathbf{x}$ -direction, so write $\mathbf{v} = v\mathbf{x}$. The $\mathbf{B}$ -field is in the $-\mathbf{z}$ -direction, so write $\mathbf{B} = -B\mathbf{z}$, where $B > 0$. Then by bilinearity of the cross product,

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B}) = q(v\mathbf{x} \times (-B\mathbf{z})) = -qvB(\mathbf{x} \times \mathbf{z}) = -qvB(-\mathbf{y}) = qvB\mathbf{y}.$$

If the charge is positive, then the force is up, and otherwise it is down. Since the force is up in the image, the charge *is positive*.

(b) Since $\mathbf{v}$ and $\mathbf{B}$ are perpendicular, the magnetic force $F_B = qvB$. The centripetal force is $F_c = \frac{mv^2}{R} = \frac{pv}{R}$. Equating these yields

$$\frac{pv}{R} = qvB \implies p = qBR.$$

To find R , we note that the circle is centered at $(0, R)$ and contains the point (a, d) . This point therefore satisfies the circle equation $x^2 + (y - R)^2 = R^2$:

$$a^2 + (d - R)^2 = R^2 \implies a^2 + d^2 - 2dR + R^2 = R^2$$

$$\implies a^2 + d^2 = 2dR.$$

We thus have

$$R = \frac{a^2 + d^2}{2d}.$$

Plugging this in yields

$$p = \boxed{\frac{qB(a^2 + d^2)}{2d}}.$$